

LUKE BUONANNO

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PROFESSIONAL PROFILE

Aspiring Mechanical Engineer combining hands-on CAD and product design experience with programming, embedded systems, and hardware troubleshooting skills. Builds independent hardware and firmware projects while applying strong analytical thinking and systematic problem-solving across multidisciplinary engineering challenges. Brings a collaborative mindset, effective communication, adaptability, and leadership experience, with a proven ability to work well on teams, resolve challenges constructively, and quickly learn new technologies.

EDUCATION

King's College, Wilkes-Barre, PA

Anticipated Graduation: 2028

Bachelor of Science in Mechanical Engineering with concentrations in Computer Science and Mathematics

GPA: 3.457

Dean's List: Fall 2024, Spring 2025

TECHNICAL SKILLS

CAD & Design: Autodesk Inventor (advanced, 4+ years, 3D design & drafting), AutoCAD, SolidWorks (working knowledge), Cura slicer

Programming: HTML5 (advanced), Discord.js / Discord.py / Node.js (advanced), Python, C++, Java (intermediate); strong comprehensive project development, debugging, and logic/systems planning across languages

Hardware & Lab: LabVIEW, Arduino IDE, multimeter/oscilloscope use, schematic reading, breadboarding, soldering

Tools & Platforms: Git, Linux, Windows, Microsoft Office, VS Code, Eclipse

Languages: Intermediate Spanish

ENTREPRENEURSHIP

Lemony Studios — Founder & Engineer

May 2024 – Present

- Design and develop embedded hardware systems centered on ESP32 and Raspberry Pi platforms, integrating power management, displays, sensors, user input, and custom electronics.
- Engineer mechanical components and functional prototypes using CAD and additive manufacturing, iterating designs from concept through testing.
- Develop firmware and software for embedded devices, including user interfaces, peripheral integration, networking, and hardware validation.
- Design and maintain a technical portfolio website documenting embedded systems, CAD designs, firmware development, and engineering projects using HTML/CSS.
- Conceptually advised a University of Pittsburgh research colleague on a diamond nitrogen-vacancy quantum sensor research setup, proposing a simplified sample-access solution to replace an existing 16-bolt assembly

PROJECTS

HenHacks Hackathon, University of Delaware, In-Person

March 2025 and 2026

- Collaborated on teams to design, build, and present two projects, “Dose Doodle” (a prescription medicine assistant) and “Hen-intersection,” (the traffic intersection reimagined for pedestrians) each within a 24-hour hackathon timeframe (<https://devpost.com/LukeBuonanno>)

PROFESSIONAL EXPERIENCE

King's College Resident Life, Resident Assistant, Wilkes-Barre, PA

August 2022 – Present

- Provide structure and support to a community of 13–25 residents, many attending college for the first time, mediating peer conflicts and enforcing Residence Life policies
- Plan and lead community programs and events that foster connection and support among residents

IITS, King's College, IT Student Support Specialist, Wilkes-Barre, PA

October 2024 – March 2025

- Resolved hardware, software, networking and printer-related issues for students and faculty while maintaining rapid response times across campus support requests.

Wegmans Food Markets, Front End Employee, Allentown, PA

August 2022 – Present

- Processed customer transactions efficiently, consistently operating the cash register approximately 20% faster than the store average while maintaining accuracy.
- Delivered exceptional customer service by providing product recommendations, resolving concerns, and earning more than 50 customer commendations from management for service quality.

St. Luke's Hospital, Volunteer Patient Advocate, Palmerton, PA

May 2022 – May 2024

- Supported patients by offering companionship and providing comfort items such as drinks and blankets upon request

ACTIVITIES, LEADERSHIP & AWARDS

- Treasurer, Engineering Club, King's College | Vice President, Coding Club, King's College